GeoGPT4V: Towards Geometric Multi-modal Large Language Models with Geometric Image Generation

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Abstract

Large language models have seen widespread adoption in math problem-solving. However, in geometry problems that usually require visual aids for better understanding, even the most advanced multi-modal models currently still face challenges in effectively using image information. High-quality data is crucial for enhancing the geometric capabilities of multi-modal models, yet existing open-source datasets and related efforts are either too challenging for direct model learning or suffer from misalignment between text and images. To overcome this issue, we introduce a novel pipeline that leverages GPT-4 and GPT-4V to generate relatively basic geometry problems with aligned text and images, facilitating model learning. We have produced a dataset of 4.9K geometry problems and combined it with 19K open-source data to form our GeoGPT4V dataset. Experimental results demonstrate that the GeoGPT4V dataset significantly improves the geometry performance of various models on the MathVista and MathVision benchmarks. The code is available at https://github.com/Lanyu0303/GeoGPT4V_Project.

1 Introduction

With large language models (LLMs) demonstrating formidable performance, their application in solving mathematical problems has become an increasingly popular trend (Toshniwal et al., 2024; Wang et al., 2023b; Gou et al., 2023; Wang et al., 2023a). Prior research has indicated that humans encounter a significant reduction in accuracy when resolving geometric problems devoid of visual aids (Chen et al., 2021). Thus, the integration of visual information from images is imperative for accurately solving of such mathematical problems, necessitating the visual perception capabilities of multi-modal large language models (MLLMs). However, even the best batch of MLLMs available now (such as GPT-4V (OpenAI, 2023b), Gemini (Anil et al., 2023)) still lag significantly behind human performance (Wang et al., 2024). Therefore, researchers are eagerly exploring methods to enhance the geometric capabilities of MLLMs.

To enhance the geometric capabilities of MLLMs, an important step is to construct corresponding high-quality data (Gao et al., 2023; Zhou et al., 2023b; Chen et al., 2022). Nevertheless, current data often suffer from two main issues. On the one hand, most open-source datasets are quite challenging, making it difficult for models to directly learn geometric capabilities from them (Bengio et al., 2009; Xu et al., 2020). For instance, the UniGEO (Chen et al., 2022) dataset consists of problems extracted from high school textbooks, but the models have not been exposed to the corresponding foundational knowledge. On the other hand, current data augmentation techniques (Gao et al., 2023), using ChatGPT-3.5 to adjust numerical values in the text, fail to harmonize these changes with the corresponding values in images. Consequently, mismatches between the altered text and images can bewilder the model and impede its learning process (Hessel et al., 2021; Yao et al., 2022).

In this paper, we address the aforementioned issues by introducing a straightforward and efficient pipeline for generating geometric problem data. Our objectives are two-fold: (1) to create geometric problems that facilitate the model's acquisition of basic geometric concepts, and (2) to ensure that the image and the text of the generated geometric problems are well-aligned. In detail, we first employ GPT-4V to create a collection of simplified geometric problems based on open-source datasets. Subsequently, we harness the capabilities of GPT-4 (OpenAI, 2023a) to generate K individ-

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ual pieces of Wolfram[1] code for each geometric problem previously crafted. The code is then executed to produce K distinct geometric images. Finally, GPT-4V is employed to score these images, allowing us to select the best one that optimally aligns with the associated textual descriptions.

Through the above pipeline, we generate a dataset comprising 4.9K geometric problems characterized by simplicity and image-text matching. We then mix our generated problems with 19K problems from open-source datasets to formulate a dataset with various difficulty levels, named GeoGPT4V. We have conducted comprehensive experiments on the geometry problem subset of MathVista (Lu et al., 2024b) and MathVision (Wang et al., 2024) datasets, two commonly used datasets for multi-modal math. Our experimental results show that models of various sizes and types can achieve significant improvements in geometric capabilities after training with our dataset (achieving 58.2% and 33.8% relative improvement for LLaVA-1.5-7B (Liu et al., 2023b) and ShareGPT4V-7B (Chen et al., 2023a), respectively, on Geometry problem solving (GPS) minitest split of MathVista), which validates the effectiveness of our approach.

In conclusion, the contributions of this paper are summarized as follows:

- We first introduce a novel pipeline capable of automatically generating simple geometric data with aligned image-text pairs.
- We have open-sourced the 4.9K dataset generated by our pipeline, along with the checkpoints of models trained on GeoGPT4V, to facilitate the community's growth and development.
- Extensive experiments have consistently shown that GeoGPT4V effectively enhances the multi-modal geometric capabilities of models of various types and sizes.

2 Related Work

In this section, we delve into related studies from two perspectives: multi-modal large language models and mathematical problem solving.

Multi-modal Large Language Models. With the rapid advancement of LLMs, the research community has started to develop multi-modal extensions of these models, known as MLLMs (Bai et al., 2023; OpenAI, 2023b; Liu et al., 2023c). These MLLMs integrate visual information with linguistic data, enhancing their capabilities significantly (Lu et al., 2024a; Li et al., 2023; Ye et al., 2023; Dai et al., 2023). Closed-source models, such as GPT-4V (OpenAI, 2023b), Gemini (Anil et al., 2023), and Qwen-VL-Max (Bai et al., 2023), have demonstrated remarkable proficiency in image comprehension and cognitive tasks. For open-source models, LLaVA (Liu et al., 2023c,b, 2024) utilizes linear projection to bridge the visual encoder and the language model, achieving commendable performance in multi-modal tasks. Building upon the LLaVA architecture, ShareGPT4V (Chen et al., 2023a) employs high-quality instructional data to further enhance model capabilities. Moreover, InternVL-Chat (Chen et al., 2023b) upscales its visual encoder to 6 billion parameters. InternLM-XComposer2 (Dong et al., 2024) excels in free-form text-image composition and understanding. Although these MLLMs have shown powerful visual capabilities, MLLMs still confront challenges when it comes to mathematical problem-solving, as highlighted by recent studies (Wang et al., 2024; Lu et al., 2024b; Yue et al., 2023).

Mathematical Problem Solving. The remarkable reasoning capabilities of LLMs have spurred researchers to harness them for solving mathematical problems (Zhou et al., 2023a; Shao et al., 2024; Lightman et al., 2023; Zhao et al., 2023). In the realm of pure text-based mathematical tasks, WizardMath (Luo et al., 2023) enhances model performance by refining instructions through a process of downward and upward instruction evolution. MetaMath (Yu et al., 2023) approaches the challenge by bootstrapping mathematical questions and rewriting them from various perspectives to improve understanding and problem-solving. However, as previous studies have found, humans' accuracy significantly decreases when solving geometry problems without images (Chen et al., 2021). Therefore, geometry problems necessitate the visual perception abilities of multi-modal models to fully comprehend and solve them. UniGeo (Chen et al., 2022) addresses this by compiling geometry problems from high school textbooks and introducing a unified multitask geometric transformer framework to tackle calculation and proving problems simultaneously in the form of sequence generation. G-LLaVA (Gao et al., 2023) leverages ChatGPT-3.5

[1]The Wolfram is a computational language designed to handle various computing and data analysis tasks, possessing a formidable capability for geometric visualization.

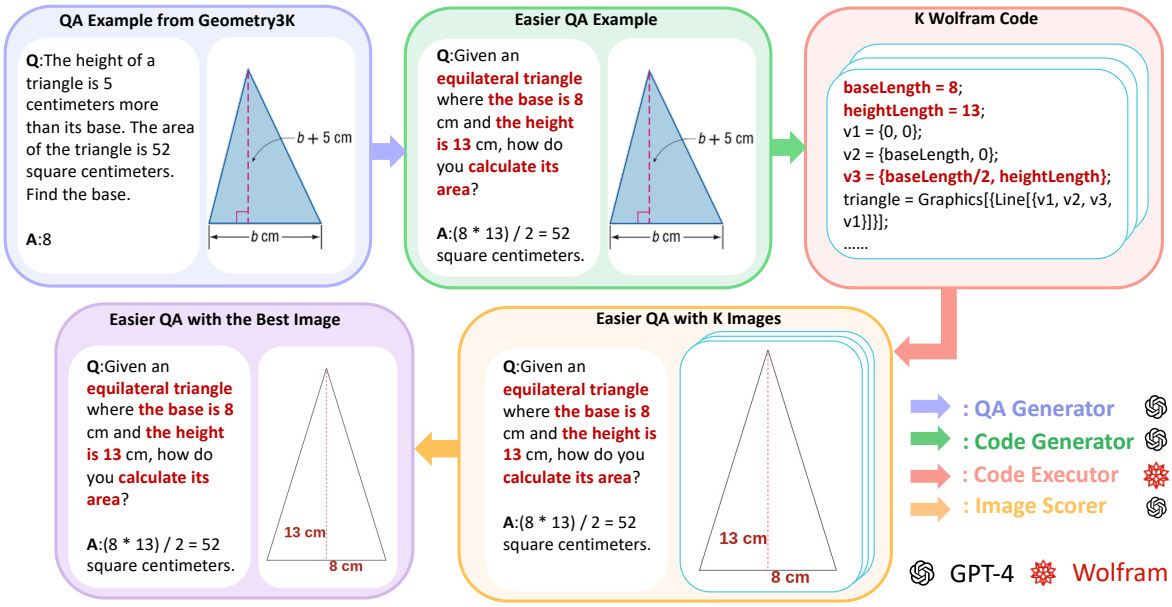


Figure 1: **Pipeline of our geometric data generation.** During the first step, we employ GPT-4V to generate simplified geometric question-answer pairs based on open-source datasets. We **highlight** the simplified parts compared to the original questions. During the second step, we employ GPT-4 to generate K Wolfram code for each question-answer pair. During the third step, we execute K code to obtain K images. During the fourth step, we employ GPT-4V to score the degree of alignment between the generated images and the questions. We choose the image with the highest score. Finally, we can obtain simplified and image-text matching geometric problems.

to create geometric question-answer pairs and to rewrite the textual content within questions. Nevertheless, this approach of textual rewriting alone may result in discrepancies between images and text, leading the model to produce incorrect or unrealistic outputs (Liu et al., 2023a). This highlights the ongoing challenge of aligning textual and visual information in multi-modal mathematical problem-solving.

3 Method

In this section, we will elaborate on the pipeline we have constructed. An overview of our pipeline is depicted in Figure 1. Specifically, our process includes: (1) generating new question-answer pairs (Section §3.1), (2) producing corresponding geometric images (Section §3.2), and (3) scoring and filtering based on the image-text matching degree (Section §3.3).

Formally, the original data from the open-source datasets can be represented as $D = \{Q, A, I\}$, where Q represents the question, A represents the answer, and I represents the image.

3.1 Question-Answer Pairs Generation

Due to the prevalence of more challenging geometric problems in open-source datasets, to facilitate our model's learning of basic geometric concepts, we initially simplify these difficult problems to generate easier geometric question-answer (QA) pairs.

In detail, we utilize GPT-4V (OpenAI, 2023b) to generate QA pairs from the dataset $D = \{Q, A, I\}$. We instruct GPT-4V to craft simplified problems that are derived from the original geometric QA pairs to acquire QA pairs containing fundamental geometric concepts. In detail, we prompt GPT-4V to consider these three perspectives: (1) generating lead-up problems, (2) generating sub-problems, and (3) incorporating the conclusions from the answer into the conditions of the question, which can reduce the complexity of the question. To prevent GPT-4V from generating the same simplified questions, we also ask GPT-4V to generate questions that are as diverse as possible. Additionally, for efficiency, the instruction also asks GPT-4V to generate textual descriptions of images aimed at supporting the subsequent phase of image generation. The detailed prompt can be found in Appendix C.1.

In practice, we generate N ($N = 3$) new data points based on a single original data point to improve efficiency and reduce API costs. After this phase, the data we obtain can be formally represented as $\tilde{D}_1 = \{\tilde{Q}, \tilde{A}, \tilde{Des}\}$ where $\tilde{Des}$ represents the image description.

3.2 Geometric Images Generation

It is important to highlight that the newly generated QA pairs may not correspond directly to the original images, which could hurt the model's learning process. To ensure congruity between the textual content and the visual aspects, it is essential to produce new images that align with the generated QA pairs. To address this issue, we employ Wolfram, a powerful software tool capable of executing code to generate geometric images.

In detail, we utilize GPT-4 (OpenAI, 2023a) to generate Wolfram code based on the dataset $\tilde{D}_1$. Firstly, we feed the questions, answers, and image descriptions as prompts to GPT-4 to generate Wolfram code. During the generation process, we instruct GPT-4 to explicitly name all variables within the code, with the aim of facilitating a clearer understanding and assisting GPT-4 in recognizing the relationships between code elements and the given questions. The detailed prompt can be found in Appendix C.2. Finally, we execute the Wolfram code, resulting in the generation of new images.

In practice, it is noticed that employing GPT-4 to generate code is unstable. Thus, we generate K ($K = 3$) distinct code from the same data to increase the probability of obtaining the correct code. Consequently, we can obtain K distinct images corresponding to K code. It can be represented as $\tilde{D}_2 = \{\tilde{Q}, \tilde{A}, \tilde{I}^{(1)}, \tilde{I}^{(2)}, \ldots, \tilde{I}^{(K)}\}$, where $\tilde{I}^{(i)}$ represents the i-th image generated for each question.

3.3 Scoring and Filtering

After generating K images using Wolfram for each question, we need to select the most suitable one to be used as the final image in our dataset.

Concretely, we employ GPT-4V to assign a score ranging from 0 to 1 that reflects the degree of correspondence between an image generated for the question and the question itself; a higher score signifies a stronger alignment. To augment the scoring proficiency of GPT-4V, drawing inspiration from the Chain-of-Thought (Wei et al., 2022) , we instruct GPT-4V to articulate the rationale underlying its evaluation before determining the ultimate score. The detailed prompt can be found in Appendix C.3.

Finally, for each question associated with K distinct generated images, we obtain K corresponding scores. For each question, we retain the image with the highest score as $\tilde{I}$. Note that, if this score is less than 0.9, we consider that the image for this question has not been well-generated, and we discard the question. Consequently, we compile a dataset $\tilde{D} = \{\tilde{Q}, \tilde{A}, \tilde{I}\}$ that consists of questions that are simpler and exhibit a stronger alignment between the images and the associated text.

4 Data Analysis

Datasets	Samples
Open-source Datasets	
ChartQA	7398
UniGEO-Calculation	3499
Geometry3K	2101
GeoQA+	6026
Generated Datasets	
UniGEO-Proving_Enhanced	1810
Geometry3K_Enhanced	1909
GeoQA_Enhanced	1212

Table 1: **The datasets used in the GeoGPT4V dataset.** Column "Samples" is the number of image-text pairs in each dataset. It is worth noting that we only use the training sets of open-source datasets.

In this section, we will present a comprehensive statistical analysis (Section §4.1) and a GPT-4V-based evaluation (Section §4.2 §4.3) of the datasets generated through our pipeline. Due to space constraints, we also present the results of the human evaluation in Appendix E

4.1 Datasets

In this study, to minimize costs, we selected the first 1,500 samples from the training sets of the UniGEO-Proving (Chen et al., 2022), Geometry3K (Lu et al., 2021), and GeoQA (Chen et al., 2021) to create UniGEO-Proving_Enhanced, Geometry3K_Enhanced, and GeoQA_Enhanced for validating the effectiveness of our method. Subsequently, we combine the generated geometric problems with those from open-source datasets, including ChartQA (Masry et al., 2022), UniGEO-Calculation (Chen et al., 2022), the original Geometry3K (Lu et al., 2021), and GeoQA+ (Cao and Xiao, 2022), to form a new dataset with various difficulty levels, dubbed GeoGPT4V. A detailed breakdown of the datasets is provided in Table 1.

4.2 Difficulty Evaluation

As mentioned in Section §3, our pipeline will take original data D as input and output generated data $\tilde{D}$. We aim to generate easier data than the original one to facilitate model learning of basic geometric knowledge. This section demonstrates the efficacy

of our pipeline by comparing the difficulty levels of D and $\tilde{D}$.

We initiate this by forming a data pair $P_1 = \{D, \tilde{D}\}$ and utilize GPT-4V to assess the relative difficulty of the data points. To mitigate the bias that GPT-4V may have due to the presentation order, we also consider the pair $P_2 = \{\tilde{D}, D\}$, obtained by swapping the order of the data points. If GPT-4V produces different outputs based on P_1 and P_2, we conclude that the difficulty of D and $\tilde{D}$ is equal. A detailed prompt can be found in Appendix C.4.

In practice, we randomly sample 500 pairs of generated and corresponding original data points. The outcome, presented in Figure 2a, reveals that over 80% of the questions in the generated dataset are of equal or lesser difficulty compared to the original questions. This indicates that our pipeline is successful in generating data that is simpler than the original dataset.

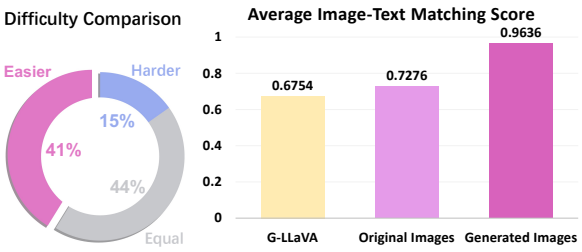
4.3 Image-text Matching Evaluation

As mentioned in the previous section, the alignment between text and images is a critical aspect of geometric problem data. To illustrate that the generated images are better suited for the simplified problems than the original images, we replace the generated images with the original image for each question, resulting in new data $\tilde{D}' = \{\tilde{Q}, \tilde{A}, I\}$. Consequently, in this section, we will compare the level of image-text matching in our generated data $\tilde{D}$ with $\tilde{D}'$ and the QA data produced by prior methods – G-LLaVA (Gao et al., 2023). Similar to the score function in Section §3.3, we employ GPT4-V to score the degree of alignment between the images and the questions.

In detail, we randomly select 500 data points for each dataset and show the average scores of the three datasets in Figure 2b. The results indicate that our generated data, $\tilde{D}$, exhibits a significantly higher degree of image-text matching than $\tilde{D}'$, as well as the dataset enhanced by G-LlaVA (0.9636 for $\tilde{D}$, 0.7276 for $\tilde{D}'$, and 0.6754 for G-LlaVA). Moreover, it is observed that G-LlaVA's image-text matching score is the lowest, which confirms our hypothesis that simply scaling the size of numbers within problems is an inappropriate approach.

5 Experiment

In this section, we conduct experiments to answer the following research questions (RQ):



(a) (b)

Figure 2: **The data analysis results.** This chart illustrates the simplicity and image-text matching attributes of our dataset. Figure (a) is a comparison chart of the difficulty between the generated and original data. In this figure, "Easier" represents that the generated data is easier than the original data; "Harder" represents that the generated data is harder than the original data; "Equal" represents that the generated and original data have the same difficulty level. Figure (b) shows the average image-text matching scores for the three data types. "Generated Images" represents our generated data. "Original Images" represents the data obtained by replacing generated images in generated data with original images.

- **RQ1**: Can GeoGPT4V dataset improve geometric capabilities of different models?
- **RQ2**: Are the generated images better than the original images for model learning?
- **RQ3**: Is it necessary to score and filter the generated images?
- **RQ4**: Is the improvement solely due to the original dataset?

5.1 Experimental Setup

Benchmarks. We utilize two widely used benchmarks, which encompass numerous multi-model geometric problems, to evaluate the effectiveness of our proposed GeoGPT4V dataset. The detailed information of these benchmarks is as follows:

- *MathVista (Lu et al., 2024b)* is a mathematical reasoning benchmark in visual contexts. It includes diverse visual contexts, such as natural images, geometry diagrams, charts, etc. MathVista includes multiple-choice questions as well as open-ended questions. The MathVista test set comprises 5141 examples without ground truth answers and provides 1000 examples with ground truth answers known as MathVista testmini.

Model	Size	MathVista			MathVision									
		GPS	GEO	AVG	AnaG	CombG	DescG	GrphT	Angle	Area	Len	SolG	TransG	AVG
LLaVA-1.5	7B	20.67*	20.92*	20.80*	7.1	7.1	7.7	10	15.6	10.2	9.8	5.3	4.8	8.62
LLaVA-1.5	13B	24.04*	23.85*	23.95*	14.3	9.1	13.5	5.6	10.4	12.6	14.7	11.5	10.7	11.38
LLaVA-1.5-G	7B	32.69	32.22	32.46	9.52	16.88	9.62	21.11	19.08	11.06	17.15	9.43	15.48	14.37
LLaVA-1.5-G	13B	36.54	37.24	36.89	15.48	14.29	12.50	18.89	19.65	13.60	18.49	9.02	11.31	15.14
ShareGPT4V	7B	21.63*	20.50*	21.07*	3.6	10.1	11.5	14.4	16.2	11.8	12.3	9.8	11.3	11.22
ShareGPT4V	13B	27.4*	27.62*	27.51*	15.5	10.7	11.5	8.9	11.6	13	17.4	10.3	12.5	12.38
ShareGPT4V-G	7B	32.69	31.80	32.25	11.90	12.99	9.62	16.67	17.34	13.60	17.59	10.25	11.31	13.47
ShareGPT4V-G	13B	43.27	42.26	42.77	22.62	9.74	13.46	11.11	19.08	15.80	13.81	9.02	13.69	14.26
InternVL†	40B	61.1	61.1	61.10	16.67*	12.99*	15.38*	13.33*	4.62*	5.60*	6.46*	9.84*	10.71*	10.62*
InternVL-G†	40B	64.42	63.60	64.01	16.67	18.18	13.46	16.67	23.12	18.40	18.93	11.89	23.21	17.84
Closed-source Models														
Qwen-VL-Plus	-	38.5	39.3	38.90	17.9	12.7	15.4	8.9	11.6	6.4	10.0	14.3	11.31	12.06
Qwen-VL-Max	-	-	-	-	19.1	16.9	16.4	12.2	13.3	14.2	19.8	11.5	17.3	15.61
Gemini-1.0-Pro	-	40.4	41.0	40.70	10.7	20.1	20.2	21.1	19.1	19.0	20.0	14.3	20.8	18.37
Gemini-1.0-Ultra	-	56.2	55.6	55.90	-	-	-	-	-	-	-	-	-	-
GPT-4V	-	50.5	51.0	50.75	32.1	21.1	22.1	14.4	22.0	22.2	20.9	23.8	25.6	22.69

Table 2: **Overall results of different models on the MathVista and MathVision.** We present the detailed scores for all the tasks related to geometry such as "GPS" and "AnaG", as well as the average score over these tasks in two benchmarks denoted as "AVG". Due to limited space, we utilize abbreviations for these geometry-related tasks and illustrate the detailed task name in the Appendix A. For the model trained with GeoGPT4V, score increases are marked in red compared to the original model. * indicates our re-implemented test results missed in benchmarks or origin papers. InternVL†represents the abbreviation for InternVL-Chat-V1.2-Plus. The suffix "-G" to the model name indicates a model trained on the GeoGPT4V. For better comparison, we also demonstrate results for five representative closed-source MLLM models.

- *MathVision (Wang et al., 2024)* is a more challenging multi-modal mathematical benchmark than MathVista. It categorizes all mathematical problems into five difficulty levels and 16 distinct tasks. MathVision also consists of multiple-choice questions and open-ended questions. The MathVision test set comprises 3040 examples with ground truth answers.

Evaluation Method. We strictly follow the evaluation method proposed in MathVista (Lu et al., 2024b) and MathVision (Wang et al., 2024). Firstly, we utilize ChatGPT-3.5 to extract the ultimate response from model outputs in MathVista, while employing regular expressions with MathVision for the same purpose. Consequently, we report the accuracy of the answers as the score for performance evaluation, with a maximum possible score of 100.

Baseline Models. We train the following mainstream open-source models using our proposed GeoGPT4V dataset, with model sizes including 7B, 13B, and 40B.

- *LLaVA-1.5 (Liu et al., 2023c,b)* utilizes linear layers to connect the vision encoder and the large language model (LLM). In the pre-training stage, LLaVA-1.5 keeps the vision encoder and the LLM frozen, and only trains linear layers. In the fine-tuning stage, it freezes the vision encoder and trains the linear layers and the LLM.

- *ShareGPT4V (Chen et al., 2023a)* has an architecture similar to LLaVA's. However, in the pre-training stage of ShareGPT4V, both the vision encoder and the language model remain unfrozen. The training data is high-quality, detailed description data generated by GPT-4V.

- *InternVL-Chat-V1.2-Plus (Chen et al., 2023b)* utilizes the InternViT (Chen et al., 2023b) as its visual encoder, which has 6 billion parameters. What's more, it scales LLM to 34B and utilizes a fine-tuning dataset with 12 million samples.

Implementation Details. For data generation, we employ the "gpt-4-vision-preview" and "gpt-4-1106-preview" API provided by OpenAI for GPT-4V and GPT-4. For model training, all the models are trained on NVIDIA A100 GPUs with PyTorch

Model	MathVista			MathVision									
	GPS	GEO	AVG	AnaG	CombG	DescG	GrphT	Angle	Area	Len	SolG	TransG	AVG
LLaVA-1.5-7B	20.67*	20.92*	20.80*	7.1	7.1	7.7	10	15.6	10.2	9.8	5.3	4.8	8.62
- Image Generation	30.77	30.96	30.87	8.33	14.94	8.65	15.56	17.34	**12.20**	14.48	7.79	**19.05**	13.15
- Image Scoring	**33.65**	31.80	**32.73**	**9.52**	15.48	9.62	20.00	17.34	**12.20**	15.59	6.56	15.48	13.54
GeoGPT4V	32.69	**32.22**	32.46	**9.52**	**16.88**	**9.62**	**21.11**	**19.08**	11.06	**17.15**	**9.43**	15.48	**14.37**

Table 3: **Ablation for image generation and image scoring.** "- Image Generation" denotes the exclusion of newly generated geometric images. "- Image Scoring" signifies the random selection of generated images, rather than utilizing GPT4V to score and choose them. For comparison, we also represent the results from the official LLaVA-1.5-7B model in the first line and GeoGPT4V in the last line. **Bold results** indicate the best results for all models. * indicates our re-implemented test results missed in benchmarks or origin papers.

version 2.0.1. To ensure a fair comparison, we keep the training parameters consistent with those specified by the model's original authors and train the models for one epoch. Detail training parameters are demonstrated in Appendix B.

5.2 Main Results (RQ1)

We evaluate the performance of various open-source models on MathVista testmini (short as MathVista) and MathVision test (short as MathVision) benchmarks after training on the GeoGPT4V dataset to demonstrate our proposed method's effectiveness. For convenience, we append the suffix "-G" to the model name to indicate a model trained on the GeoGPT4V dataset, such as "LLaVA-1.5-G". Since our method focuses on geometric data, we present detailed scores for all the tasks related to geometry and the average score over these tasks in Table 2. The complete set of scores can be found in Appendix D.1 and D.2. In Appendix D.3, we compare the geometric capabilities of our best model, InternVL-Chat-V1.2-Plus-GeoGPT4V, with other open-source and closed-source models.

The experimental results from Table 2 indicate that our dataset can effectively improve different models' geometric capabilities. First of all, our proposed GeoGPT4V has exhibited an improvement in the average scores across all geometry-related tasks on both MathVista and MathVision benchmarks, indicating that GeoGPT4V can enhance the model's general geometry performance. Moreover, our proposed GeoGPT4V has brought improvements to most geometry-related tasks in both benchmarks in all scales and types of models. Furthermore, our GeoGPT4V significantly bridges the gap in geometric capabilities between open-source and closed-source models, except InternVL-Chat-V1.2-Plus, which has already employed a substantial amount of customized fine-tuning datasets.

5.3 In-depth Analysis

To comprehensively analyze the effectiveness of GeoGPT4V, we design a series of analyzing experiments from various perspectives. Firstly, we design ablation experiments from the standpoint of the efficacy of generating new geometric images and selecting generated images with GPT4V scores. Subsequently, we conduct experiments to demonstrate the substantial performance improvement brought by GeoGPT4V stemming from the generated data rather than the utilization of open-source data. Due to resource and space limitations, we leverage LLaVA-1.5-7B for analytical experiments and conduct evaluations on both MathVista and MathVision.

5.3.1 Effect of Generating New Images (RQ2)

We validate the effectiveness of the newly generated geometric images by replacing the images generated in GeoGPT4V with their original counterparts and training the model on them. In detail, we first substitute the newly generated images from GeoGPT4V with the original images while retaining the simplified questions generated, formulating a new dataset denoted as $\tilde{D}'$. Subsequently, we train the LLaVA-1.5-7B model on $\tilde{D}'$ and compare its geometric capabilities with the model trained on GeoGPT4V.

Based on results demonstrated in Table 3, we have following observations: Firstly, the model trained on $\tilde{D}'$ exhibits inferior performance compared to the model trained on GeoGPT4V, indicating the effectiveness of the newly generated images. Secondly, the model trained on $\tilde{D}'$ demonstrates stronger performance than the model trained without the use of $\tilde{D}'$, thereby validating the efficacy of

Name	Base	Replace	Mix
Datasets	ChartQA	ChartQA	ChartQA
	UniGeo-Calculation	UniGeo-Calculation	UniGeo-Calculation
	Geometry3K	Geometry3K_Replace	Geometry3K_Mix
	GeoQA+	GeoQA+_Replace	GeoQA+_Mix
	UniGeo-Proving	UniGeo-Proving_Replace	UniGeo-Proving_Mix

Table 4: **Dataset settings for experiments comparing open-source data and generated data.** The suffix "Replace" indicates that we replace the corresponding original data with generated data. The suffix "Mix" indicates that we mix the original data with generated data.

Datasets	MathVista			MathVision								
	GPS	GEO	AVG	AnaG	CombG	DescG	GrphT	Angle	Area	Len	SolG	TransG	AVG
Base	29.33	28.03	28.68	10.71	**15.91**	8.65	12.22	16.67	11.80	13.59	8.20	16.07	12.65
Replace	33.17	32.64	32.91	7.14	14.94	6.73	**20.00**	**20.81**	10.80	**15.14**	**10.25**	14.29	13.34
Mix	**33.52**	**34.31**	**33.92**	**11.90**	15.58	**10.58**	14.44	17.34	**12.40**	14.48	9.43	**16.07**	**13.58**

Table 5: **Comparison of the effects with and without using the generated datasets. Bold results** indicate the best results for all models.

the easier QA pairs generated by our pipeline.

5.3.2 Is Scoring Necessary? (RQ3)

As mentioned in Section §3.3, K images are scored, and the one with the highest score is selected from this set. To demonstrate the necessity of scoring, we formulate a new dataset $\tilde{D}''$ by directly modifying the selection method to randomly choose from the K images while keeping all other aspects unchanged. Consequently, we analyze the performance of the LLaVA-1.5-7B trained on $\tilde{D}''$.

According to results demonstrated in Table 3, we can find that the model trained on $\tilde{D}''$ exhibits inferior performance on most tasks compared to the model trained on GeoGPT4V. The results indicate that the quality of the images obtained via ranking surpasses those chosen randomly in overall aspects.. It is also worth noting that the model trained on $\tilde{D}''$ performs better on a few tasks, possibly due to the relative similarity of the generated images in these tasks. While using GPT-4V for selection may introduce bias, random selection has the potential to enhance diversity.

5.3.3 Are the Open-source Datasets Enough? (RQ4)

To demonstrate that the performance improvements brought by GeoGPT4V are not solely reliant on open-source data, we compare the performance of models trained using various combinations of open-source and our generated data. In detail, as illustrated in Table 4, we construct three tiers of datasets. Firstly, we combine all open-source datasets to create the "Base" dataset. Subsequently, we replace the original data from the "Base" dataset with the data generated by our pipeline, resulting in the "Replace" dataset. Lastly, we mix the generated data with all the data from the "Base" dataset to form the "Mix" dataset. It is notable that GeoQA is a subset of GeoQA+. Thus we only use GeoQA+ in these three dataset settings, rather than using both GeoQA+ and GeoQA.

We finetune LLaVA-1.5-7B separately on these three datasets and evaluate their performance in Table 5, with observations as follows: Although the "Base" dataset, constructed using open-source data, provides moderate geometric capabilities, our "Replace" and "Mix" datasets exhibit even greater enhancements in geometric performance. This not only demonstrates the effectiveness of the data generated by our pipeline but also indicates that the improvements afforded by GeoGPT4V are not solely derived from open-source data.

6 Conclusion

In this study, we propose a novel pipeline to enhance the geometric capabilities of MLLMs. We have proposed data generation methods for multimodal geometric tasks involving problem simplification and the generation of images that match newly generated text. Specifically, we use GPT4V and GPT4 to generate sub-problems or lead-up problems for given geometric tasks, along with the corresponding Wolfram code that can be executed to generate geometric images. Based on

the pipeline, we have generated 4.9K simplified and image-text matching geometric problems. We mix our generated data with 19K open-source data to formulate a dataset with various difficulty levels, named GeoGPT4V. After training on the GeoGPT4V dataset, various models have improved geometric scores on both MathVista and MathVision benchmarks. The extensive experimental results demonstrate the effectiveness of the GeoGPT4V dataset. We have open-sourced the GeoGPT4V dataset and the checkpoints of models trained on the GeoGPT4V dataset, with the aim of fostering the community's growth.

Limitations

This paper focuses on the generation of geometric images. We employ GPT-4 to generate Wolfram code, which can be executed to generate images. However, this approach is unstable and may result in poor image quality. That's why we use GPT-4V to score the images, which leads to more API calls and increased costs.

What's more, this paper only considers simplifying open-source geometric problems. However, generating more complex problems is also worth considering, as it will generate more complex geometric images and help models improve complex reasoning capabilities. Our future work will explore the more accurate generation of complex geometric images.

Finally, multi-modal mathematics is not limited to geometric problems. It also includes tasks such as chart question answering and function question answering. Generating richer charts and function images is also part of our future exploration work.

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A Detailed Task Information

Table 6 shows the correspondence between abbreviations and detailed task names.

B Training Parameters

We keep the same parameters as those specified by the model's original authors. Detail parameters are shown in Table 7.

C Prompts

C.1 Prompt for Question-Answer Pairs Generation

Table 8 shows the prompt for question-answer pairs generation. We prompt GPT-4V to generate simplified geometric problems based on the open-source datasets.

C.2 Prompt for Wolfram Code Generation

Table 9 shows the prompt for Wolfram code generation. We prompt GPT-4 to generate the Wolfram code based on the information from the question, the answer, and the image description.

C.3 Prompt for Scoring

Table 10 shows the prompt for scoring. We prompt GPT-4V to score the degree of alignment between the images and the questions.

C.4 Prompt for Difficulty Comparison

Table 11 shows the prompt for difficulty comparison. We employ GPT-4V to determine which of the two problems is more difficult.

D Detailed Evaluation Results

D.1 MathVista Results

We show full MathVista testmini results in Table 12. Although our method focuses on geometric problems, the GeoGPT4V dataset can still improve the overall scores of various models, except InternVL-Chat-V1.2-Plus, which has already employed a customized fine-tuning dataset with 12 million samples.

D.2 MathVision Results

We show full MathVision test results in Table 13. We can find that the GeoGPT4V dataset can improve the scores of most tasks on MathVision for various models. The results demonstrate the effectiveness of the GeoGPT4V dataset.

D.3 Comparison with Other Models

We compare the performance of our best model, InternVL-Chat-V1.2-Plus-GeoGPT4V, with other open-source and closed-source models regarding geometric capabilities. Detailed results are in Table 14.

For MathVista, our best model achieves the best geometric scores among all models. For MathVision, our best model achieves the highest scores for average score and most geometric scores among open-source models. The experimental results demonstrate the effectiveness of the GeoGPT4V dataset.

E Human Evaluation of the Generated Data

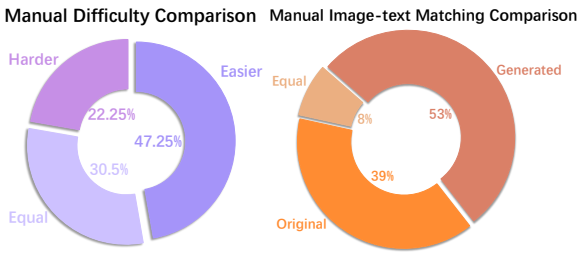
In addition to using GPT-4V to evaluate the data we generated, we hired two annotators with sufficient professional knowledge to manually evaluate the data we generated. The following are the evaluation results for difficulty and image-text matching.

E.1 Difficulty Comparison

We randomly selected 200 generated questions and their corresponding original questions, and asked annotators to compare the difficulty between the generated question and the original question. We display the results in Figure 3a, and the inner agreement between the two annotators is 0.74. In the Figure, "Easier" indicates that the generated question is easier than the original question, with other symbols following the same pattern. Based on the experimental results, 77.75% of the generated questions are easier or of the same difficulty as the original ones, which indicates that our pipeline can reduce the difficulty of the questions.

E.2 Image-text Matching Comparison

We randomly selected 200 generated questions and their corresponding original images, and asked annotators to judge which image, the generated one or the original one, better matches the generated question. We display the results in Figure 3b, and the inner agreement between the two annotators is 0.78. In the Figure, "Original" indicates that the original image better matches the question text, with other symbols following the same pattern. Based on the experimental results, we can observe that the generated images match the generated questions better than the original images.



(a) (b)

Figure 3: **The manual data analysis results.** Figure (a) is a manual comparison chart of the difficulty between the generated and original data. In this figure, “Easier” represents that the generated data is easier than the original data; “Harder” represents that the generated data is harder than the original data; “Equal” represents that the generated and original data have the same difficulty level. Figure (b) is a manual comparison chart of the image-text matching between the generated and original images. In this figure, “Original” represents that the original image better matches the question text; “Generated” represents that the generated image better matches the question text; “Equal” represents that the generated image and the original image match the text to the same degree.

Abbreviation	Task
MathVista	
FQA	Figure Question Answering
GPS	Geometry Problem Solving
MWP	Math Word Problem
TQA	Textbook question answering
VQA	Visual Question Answering
ALG	Algebraic Reasoning
ARI	Arithmetic Reasoning
GEO	Geometry Reasoning
LOG	Logical Reasoning
NUM	Numeric Commonsense
SCI	Scientific Reasoning
STA	Statistical Reasoning
MathVision	
Alg	Algebra
AnaG	Analytic Geometry
Ari	Arithmetic
CombG	Combinatorial Geometry
Comb	Combinatorics
Cnt	Counting
DescG	Descriptive Geometry
GrphT	Graph Theory
Log	Logic
Angle	Metric Geometry - Angle
Area	Metric Geometry - Area
Len	Metric Geometry - Length
SolG	Solid Geometry
Stat	Statistics
Topo	Topology
TransG	Transformation Geometry

Table 6: **Correspondence between abbreviations and detailed task names in MathVista and MathVision benchmarks.**

Parameters	LLaVA-1.5	ShareGPT4V	InternVL-Chat-V1.2-Plus
Train Epochs	1	1	1
Global Batch Size	128	128	128
Learning Rate	$2e^{-5}$	$2e^{-5}$	$1e^{-5}$
Learning Rate Schedule	cosine decay	cosine decay	cosine decay
Weight Decay	0	0	0.05
Warmup Ratio	0.03	0.03	0.03
Optimizer	AdamW	AdamW	AdamW
Tune Visual Encoder	False	False	False
Tune MLP	True	True	True
Tune LLM	True	True	True

Table 7: **Training parameters of different models.** To make a fair comparison, we keep the training parameters consistent with those specified by the model's original authors and train the models for one epoch.

Please act as a question generator.
Give you a question and its answer, along with a corresponding image for the question; please generate new questions and provide new answers in English. The new questions and new answers must meet the following conditions:
1. The new questions are slightly easier than the original ones but shouldn't be too simple.
2. Do not merely rephrase the question; you must reduce its difficulty level.
3. The new question must include a detailed description of the information in the image, which must be detailed enough to allow others to redraw the image based on the description.
5. The questions should be as diverse as possible.
6. The new answers must be correct.
Some useful tips:
1. You can incorporate information from the original answer into the question.
3. You can generate lead-up problems for the original problem.
5. You can generate sub-problems for the original problem.
4. Imagine that others cannot see the image corresponding to the new question; you must describe it using words.
5. For each question, consider it as a standalone item. Others can only view one question at a time, so avoid using phrases like "similar to the previous question" or references such as "New_Image 1".
Come up with three diverse questions and answers.
Input format:
Question: <question example>
Answer: <answer example>
You must follow this output format:
New_Question: <new question example>
New_Answer: <new answer example>
Image_Description: <new image description example>

Table 8: **Prompt for Question-Answer Pairs Generation.** We prompt GPT-4V to generate simplified questions. We also prompt GPT-4V to generate questions that are as diverse as possible to prevent GPT-4V from generating the same questions.

You are a teacher creating an exam, and you need to draw images for the questions on the exam.
Give you a question, an answer, and an image description, and generate the image corresponding to the question using Mathematica code. Your code must meet the following conditions:
1. Only use the "Export" command at the end of the code to save the generated image to "/temp/image.png".
2. The image should be clear and correspond to the question, with particular attention to shape and angle.
3. You only need to generate the image; there is no need to solve the problem.
4. All variables in the code should be named for easy understanding; avoid using terms such as "C" directly.
Some useful tips:
1. Focus on the image description.
2. You can use the information from the question and answer to help you generate code.
Come up with one code.
Input format:
Question: <question example>
Answer: <answer example>
Image description: <image description example>
You must follow this output format:
Code: <code example>

Table 9: **Prompt for Wolfram Code Generation.** When prompting GPT-4, we integrate both image descriptions and question-answer data to refine code generation. Additionally, we prompt GPT-4 to ensure variable naming within the code for clarity, aiming to enhance GPT-4's grasp of the code's relationship to the query at hand.

Please act as a scorer.
Give you a description, along with an image. Please evaluate the degree of match between the image and the description and give a score. The evaluation process must meet the following conditions:
1. The score is a decimal between 0 and 1.
2. The score reflects the degree of image-description match.
3. If the image and the image description do not match, the score should be low.
4. The score should be lower if the image is not clear enough or difficult to understand.
5. The image should be rated low if it contains only text and numbers, with no geometric shapes or chart forms.
6. The image must have clear shapes and labels.
Some useful tips:
1. Don't always give high scores.
2. Only give high scores when the image and the description match very well.
3. You can use two decimal places to represent your score.
Come up with one score.
Input format:
Image description: <image description example>
You must follow this output format:
Reason: <your reason example>
Score: <score example>

Table 10: **Prompt for Scoring.** We employ GPT-4V to score the degree of alignment between the generated images and the questions. Specifically, the score is a decimal that ranges from 0 to 1. We also prompt GPT-4V to give a reason first and then give a final score, hoping this can enhance the accuracy of scoring.

Please act as a difficulty level evaluator.
Give two geometric data, each consisting of a question, an answer, and an image.
Please compare these two questions to determine which one is more difficult.
If the first one is more difficult, output "1"; if the second one is more difficult, output "2".
Some useful tips:
1. You should consider the complexity and difficulty of the questions and images.
2. Don't automatically assume that multiple-choice questions are easier.
3. A shorter answer does not mean it's easier.
Input format:
Question_1: <the first question>
Answer_1: <the first answer>
Question_2: <the second question>
Answer_2: <the second answer>
The first image corresponds to the first question, and the second image corresponds to the second question.
You can only output the number "1" or "2".

Table 11: **Prompt for Difficulty Comparison.** We prompt GPT-4V to determine which of the two questions is more difficult. We instruct GPT-4V not to simplistically assume that multiple-choice questions or shorter answers imply an easier question.

Model	Size	All	FQA	GPS	MWP	TQA	VQA	ALG	ARI	GEO	LOG	NUM	SCI	STA
LLaVA-1.5	7B	25.1*	23.79*	20.67*	12.90*	39.24*	32.40*	24.20*	22.10*	20.92*	16.22*	18.75*	36.89*	22.26*
LLaVA-1.5	13B	27.3*	22.68*	24.04*	16.67*	42.41*	35.75*	27.40*	24.93*	23.85*	18.92*	25.00*	39.34*	22.59*
LLaVA-1.5-G	7B	30.7	28.25	32.69	18.28	42.41	34.64	32.38	25.78	32.22	32.43	23.61	42.62	26.58
LLaVA-1.5-G	13B	32.2	28.25	36.54	19.89	41.14	37.99	35.23	28.05	37.24	27.03	26.39	42.62	27.57
ShareGPT4V	7B	27.3*	21.93*	21.63*	19.35*	43.04*	36.31*	24.91*	27.20*	20.50*	18.92*	22.92*	40.16*	21.93*
ShareGPT4V	13B	30.4*	23.97*	27.40*	25.81*	43.67*	36.87*	28.83*	31.16*	27.62*	10.81*	26.39*	41.80*	26.91*
ShareGPT4V-G	7B	30.4	26.77	32.69	20.97	40.51	34.08	31.67	26.91	31.80	21.62	20.83	40.98	25.52
ShareGPT4V-G	13B	34.1	27.51	43.27	23.12	43.04	36.87	39.86	29.18	42.26	27.03	24.31	44.26	27.57
InternVL†	40B	59.9	51.7	61.1	79.6	52.5	57.0	54.5	63.2	61.1	16.2	48.6	55.7	60.8
InternVL-G†	40B	56.2	46.10	64.42	75.27	51.90	45.81	57.30	54.96	63.60	18.92	39.58	53.28	55.81

Table 12: **Overall results of different models on the MathVista.** For the model trained with GeoGPT4V, score increases are marked in red compared to the original model. * indicates our re-implemented test results missed in benchmarks or origin papers. InternVL†represents the abbreviation for InternVL-Chat-V1.2-Plus. The suffix "-G" to the model name indicates a model trained on the GeoGPT4V. We present the detailed score for all the tasks such as "FQA" and "GPS", as well as the overall (All) score for the benchmark. Due to limited space, we utilize abbreviations for the tasks and illustrate the detailed task name in the Appendix A.

Model	Size	All	Alg	AnaG	Ari	CombG	Comb	Cnt	DescG	GrphT	Log	Angle	Area	Len	SolG	Stat	Topo	TransG
LLaVA-1.5	7B	8.52	7.0	7.1	10.7	7.1	4.8	10.5	7.7	10.0	9.2	15.6	10.2	9.8	5.3	8.6	4.4	4.8
LLaVA-1.5	13B	11.12	7.0	14.3	14.3	9.1	6.6	6.0	13.5	5.6	13.5	10.4	12.6	14.7	11.5	13.8	13.0	10.7
LLaVA-1.5-G	7B	12.89	8.41	9.52	9.29	16.88	6.55	10.45	9.62	21.11	12.61	19.08	11.06	17.15	9.43	13.79	13.04	15.48
LLaVA-1.5-G	13B	13.98	9.28	15.48	16.43	14.29	10.71	10.45	12.50	18.89	11.76	19.65	13.6	18.49	10.25	13.79	17.39	13.10
ShareGPT4V	7B	10.53	5.5	3.6	12.9	10.1	4.8	7.5	11.5	14.4	10.9	16.2	11.8	12.3	9.8	15.5	17.4	11.3
ShareGPT4V	13B	11.88	7.5	15.5	16.4	10.7	8.9	9.0	11.5	8.9	7.6	11.6	13.0	17.4	10.3	8.6	8.7	12.5
ShareGPT4V-G	7B	12.80	7.83	11.9	15.00	12.99	5.95	7.46	9.62	16.67	15.97	17.34	13.60	17.59	10.25	15.52	8.70	11.31
ShareGPT4V-G	13B	12.63	8.41	22.62	15.00	9.74	6.55	8.96	13.46	11.11	15.13	19.08	15.80	13.81	9.02	6.90	13.04	13.69
InternVL†	40B	9.18*	8.41*	16.67*	8.57*	12.99*	9.52*	10.45*	15.38*	13.33*	11.76*	4.62*	5.60*	6.46*	9.84*	12.07*	21.74*	10.71*
InternVL-G†	40B	16.12	9.57	16.67	15.00	18.18	10.71	10.45	13.46	16.67	16.81	23.12	18.4	18.93	11.89	6.90	13.04	23.21

Table 13: **Overall results of different models on the MathVision.** For the model trained with GeoGPT4V, score increases are marked in red compared to the original model. * indicates our re-implemented test results missed in benchmarks or origin papers. InternVL†represents the abbreviation for InternVL-Chat-V1.2-Plus. The suffix "-G" to the model name indicates a model trained on the GeoGPT4V. We present the detailed score for all the tasks such as "Alg" and "AnaG", as well as the overall (All) score for the benchmark. Due to limited space, we utilize abbreviations for the tasks and illustrate the detailed task name in the Appendix A.

Model	Size	MathVista			MathVision									
		GPS	GEO	AVG	AnaG	CombG	DescG	GrphT	Angle	Area	Len	SolG	TransG	AVG
InternVL-G†	40B	**64.42**	**63.6**	**64.01**	16.67	18.18	13.46	16.67	**23.12**	18.40	18.93	11.89	23.21	17.84
Open-source Models														
LLaVA-1.5	13B	24.04*	23.85*	23.95*	14.3	9.1	13.5	5.6	10.4	12.6	14.7	11.5	10.7	11.38
ShareGPT4V	13B	27.4*	27.62*	27.51*	15.5	10.7	11.5	8.9	11.6	13	17.4	10.3	12.5	12.38
G-LLaVA‡	13B	56.25*	51.88*	54.07*	9.52*	7.79*	8.65*	7.78*	8.67*	12.20*	10.02*	7.38*	8.93*	8.99*
InternLM-VL†	7B	63.0	62.3	62.65	15.5	15.3	14.4	**22.2**	19.7	15.6	15.0	11.9	15.5	16.12
InternVL†	40B	61.1	61.1	61.1	16.67*	12.99*	15.38*	13.33*	4.62*	5.60*	6.46*	9.84*	10.71*	10.62*
Closed-source Models														
Qwen-VL-Plus	-	38.5	39.3	38.90	17.9	12.7	15.4	8.9	11.6	6.4	10.0	14.3	11.31	12.06
Qwen-VL-Max	-	-	-	-	19.1	16.9	16.4	12.2	13.3	14.2	19.8	11.5	17.3	15.61
Gemini-1.0-Pro	-	40.4	41.0	40.70	10.7	20.1	20.2	21.1	19.1	19.0	20.0	14.3	20.8	18.37
Gemini-1.0-Ultra	-	56.2	55.6	55.90	-	-	-	-	-	-	-	-	-	-
GPT-4V	-	50.5	51.0	50.75	**32.1**	**21.1**	**22.1**	14.4	22.0	**22.2**	**20.9**	**23.8**	**25.6**	**22.69**

Table 14: **Overall results of our best model and other open-source and closed-source models on the MathVista and MathVision.** We present the detailed score for all the tasks related to geometry such as "GPS" and "AnaG", as well as the average score over these tasks in two benchmarks denoted as "AVG". Due to limited space, we utilize abbreviations for these geometry-related tasks and illustrate the detailed task name in the Appendix A. **Bold results** indicate the best results for all models, and the red results indicate the best results among the open-source models. ‡indicates our re-implemented model without an official checkpoint. * indicates our re-implemented test results missed in benchmarks or origin papers. InternVL†represents the abbreviation for InternVL-Chat-V1.2-Plus. InternLM-VL†represents the abbreviation for InternLM-XComposer2-VL. The suffix "-G" to the model name indicates a model trained on the GeoGPT4V.